

Bulk RNA-Seq

Construct co-expression network:

Motive: Select gene-to-gene interaction network method including Distance to Empirical Measure, Pearson Correlation, Distance Correlation and Weighted Overlapping Measures



Perform persistence homology on co-expression network:

Motive: Identify topological features and filter most persistent features for biological importance



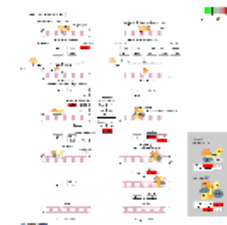
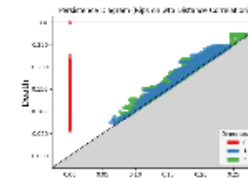
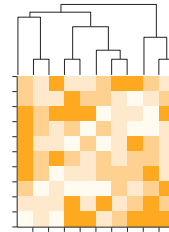
Extract birth-death intervals and co-occurrence network

Motive: Visualise the biological network, obtain network statistics and identify hub genes



Biomarker Discovery

Motive: Gain biological insights from co-occurrence network and hub genes using KEGG enrichment analysis



Construct persistence landscapes:

Motive: Vectorize persistence features to feed into ML models



Patient-level classification:

Motive: Enable data-driven insights and predictions

